Lara Koshal

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Education

York University Toronto, ON

Bachelor of Science in Computer Science (Honors) *Expected: December 2027*

Experience

Software Developer Co-op September 2026 - Present

Ontario Public Service *North York, ON*

- Developed **Playwright** and **TypeScript** scripts to automate U**I regression testing** for the VQRP application, covering end-to-end workflows for forms, navigation, search, and approvals.
- Supported QA coverage and test consistency across **100+ test cases and workflows** by verifying user inputs, expected results, and application behaviour in a dedicated QA environment, while documenting defects for investigation.

Programmer Co-op January 2025 - December 2025

Ontario Public Service *North York, ON*

- Built and maintained internal government forms with **TypeScript, React, and the Ontario Design System** to migrate workflows from Microsoft InfoPath to modern web applications; integrated applications with **SharePoint** for data management across **5+ teams/divisions.**
- Designed and maintained **100+ automated and manual test cases** using **Python, Robot Framework, and Selenium**, covering functional, regression, and edge-case scenarios across government applications.
- Automated and executed **regression and performance test suites** in an **Agile Scrum** environment using **Git and Azure DevOps**, identifying defects, validating fixes, and supporting production stability across sprint releases.

Tutor August 2024 – Present

MegaMind Learning Center *Brampton, ON*

- Tutor students one-on-one and in small groups in Math, English, and introductory programming (Scratch, Python), explaining concepts clearly and adapting lessons to each student's level

Projects

Electric Vehicles E-Commerce Platform

- Engineered a full-stack EV e-commerce platform with **7+ RESTful services using Jakarta EE, JAX-RS, and Maven**, supporting vehicle search, filtering, shopping cart, and checkout workflows.
- Improved application security and reliability through **DAO-based MariaDB operations, JUnit cart tests, password hashing, and parameterized SQL queries**; deployed services using **AWS Elastic Beanstalk.**

Role-Based Mentorship Matching Platform - Ellehacks 2026

- Built a full-stack mentorship platform at ElleHacks 2026 using **React and Supabase**, supporting 2 user roles and testing application workflows with **50+ user profiles.**
- Secured user data by implementing **PostgreSQL Row-Level Security (RLS)** and authentication-based permissions to restrict access to authorized records.

AI Policy Assistant Chatbot

- Developed a **Retrieval-Augmented Generation (RAG) pipeline** across **10+ government policy documents** using text embeddings and vector similarity search to retrieve relevant passages and generate contextual, policy-grounded **LLM responses.**
- Evaluated retrieval relevance and LLM-generated responses across **20+ test queries and edge cases**, testing answer grounding, document relevance, and potentially inaccurate outputs; implemented **authenticated test users, role-based access controls, and session handling** to validate secure interactions and access permissions.

Technical Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, C, C#
Libraries and Framework : React, Playwright, Selenium, Robot Framework, Node.js, Pandas, Numpy
Tools: MariaDB, PostgreSQL, AWS, REST, Git, Azure DevOps, Linux, SharePoint